

Introduction

Baroid Core Homework 11

Assessment Feedback

Congratulations, you passed.

Total score: 62 out of 62, 100%

Question Feedback

Question

1 of 30

A contaminant is defined as: _____

- ☐ any fluid additives that alters the physical and chemical properties of a drilling fluid
- ☐ any influx of different materials that changes the formulation of a drilling fluid
- ☒ any undesirable component that causes a detrimental effect when incorporated in a drilling fluid
- ☐ any desirable component that causes a beneficial effect when incorporated in a drilling fluid

1 out of 1

Question

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The primary indication of contamination is a change in: _____

- ☐ Drilling fluid cost
- ☐ Drilling operations
- ☒ Drilling fluid properties
- ☐ All of the above

1 out of 1

Question

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The most common contaminants encountered in the drilling fluid systems while drilling are: _____

- ☒ Hydrocarbons
- ☐ Bacteria
- ☒ Drilled solids
- ☐ Fluid additives
- ☐ Inert gases
- ☒ Evaporite salts
- ☐ Heavy metals
- ☒ Water flows
- ☒ Acid gases

5 out of 5

Question

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Severity of the problems experienced in the drilling fluids related to the contaminants depends of: _____

- ☐ Type of contaminant
- ☐ Quantity of contaminants
- ☐ Type and composition of the drilling fluid
- ☒ All of the above

1 out of 1

Question

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The drilled solids contaminants are classified into two (2) categories - **active** solids (such as **clays and shales**) and **inactive** solids (such as **calcium carbonate and sand**) .

4 out of 4

Question

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What are the most common Acid Gases that we encountered while drilling? _____

- ☒ Carbon dioxide, CO₂
- ☐ Hydrogen, H₂
- ☒ Hydrogen sulfide, H₂S
- ☐ Answers a and b
- ☐ All of the above

2 out of 2

Question

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Pilot test can be defined as: _____

- ☐ A test carried out in the environment in which a product or fluid system is to be used
- ☒ A preliminary assessment on a small volume of fluid under controlled conditions in the lab
- ☐ An early assessment of the product's effect on the circulating volume in the field
- ☐ A test that is done in the lab where the appropriate equipment, supplies, and certified expertise are available

1 out of 1

Question

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What is the formula for bicarbonate of soda? _____

- ☐ Na_2HCO_3
- ☒ NaHCO_3
- ☐ NaHCO_2
- ☐ Na_2HCO_2

1 out of 1

Question

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What is the function of bicarbonate of soda? _____

- ☒ Treat out cement
- ☐ Treat out CO_2
- ☐ Suspension agent
- ☐ Alkalinity source

1 out of 1

Question

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What is the formula for lime? _____

- ☐ $\text{Ca}_2(\text{OH})_2$
- ☒ $\text{Ca}(\text{OH})_2$
- ☐ CaOH
- ☐ Ca_2OH

1 out of 1

Question

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What is the formula of soda ash?

- ☐ Na_2CO_2
- ☐ NaCO_3
- ☒ Na_2CO_3
- ☐ NaCO_2

1 out of 1

Question

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What is the function of soda ash?

- ☒ Treat out calcium
- ☐ Treat out sodium
- ☐ Treat out CO_2
- ☐ Weighting agent

1 out of 1

Question

13 of 30

What is the formula for Caustic soda?

- ☒ NaOH
- ☐ $\text{Na}_2(\text{OH})$
- ☐ $\text{Na}_2(\text{OH})_2$
- ☐ $\text{Na}(\text{OH})_2$

1 out of 1

Question

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What is the function of Caustic soda?

- ☐ Filtration control
- ☐ Thinner

- ☒ Alkalinity source
- ☐ Shale stabilizer

1 out of 1

Question

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What is the formula for Gypsum?

- ☐ Ca_2SO_4
- ☐ CaSO
- ☐ CaSO_3
- ☒ $\text{CaSO}_4 : 2\text{H}_2\text{O}$

1 out of 1

Question

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What is the function of Gypsum?

- ☒ Treat out carbonates
- ☐ Treat out cement
- ☐ Viscosifier
- ☐ Calcium source

1 out of 1

Question

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In Water based fluids, when performing filtrate analysis we have the following alkalinity ions:

- When $2\text{Pf} > \text{Mf}$ we have: Hydroxides and Carbonates ▼
- When $2\text{Pf} < \text{Mf}$ we have: Carbonates and Bicarbonates ▼
- When $\text{Pf} = 0$ we have: Bicarbonates only ▼
- When $2\text{Pf} = \text{Mf}$ we have: Carbonates only ▼
- When $\text{Pf} = \text{Mf}$ we have: Hydroxides only ▼

5 out of 5

Ratio	OH	CO ₃	HCO ₃
Pf = Mf	✓	0	0
2Pf > Mf	✓	✓	0
2Pf = Mf	0	✓	0
2Pf < Mf	0	✓	✓
Pf = 0	0	0	✓

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	13.7
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	17
Retort Oil/Water	0 / 83	0 / 83
MBT	2.5	2.5
pH	10.0	8.3
Pm	1.5	0.4
Pf/Mf	1.0/1.8	0.1/0.2
Chlorides	300	300
Hardness	100	0

Which of the following is the contamination?

Hydrogen sulfide gas ▼

Which of the following is used as the primary treatment for this contamination?

H2S Scavenger ▼

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	185
Weight	14.0	14.0
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	17
Retort Oil/Water	0 / 83	0 / 83
MBT	2.5	2.5
pH	10.0	9.1
Pm	1.5	0.7
Pf/Mf	1.0/1.8	0.7/1.1
Chlorides	300	300
Hardness	100	100

Which of the following is the contamination?

High temperature ▼

Which of the following is used as the primary treatment for this contamination?

Dilution ▼

2 out of 2

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	14.1
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	19
Retort Oil/Water	0 / 83	0 / 81
MBT	2.5	2.5
pH	10.0	9.9
Pm	1.5	1.3
Pf/Mf	1.0/1.8	0.7/1.4
Chlorides	300	11000
Hardness	100	150

Which of the following is the contamination?

Salt ▼

Which of the following is used as the primary treatment for this contamination?

Dilution ▼

2 out of 2

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	14.1
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	19
Retort Oil/Water	0 / 83	0 / 81
MBT	2.5	2.5
pH	10.0	8.3
Pm	1.5	0.5
Pf/Mf	1.0/1.8	0.2/1.3
Chlorides	300	300
Hardness	100	800

Which of the following is the contamination?

Anhydrite 

Which of the following is used as the primary treatment for this contamination?

Soda ash 

2 out of 2

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	13.6
Vis	45	60
PV	30	27
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	13
Retort Oil/Water	0 / 83	0 / 87
MBT	2.5	2.0
pH	10.0	8.3
Pm	1.5	0.5
Pf/Mf	1.0/1.8	0.2/1.3
Chlorides	300	6000
Hardness	100	150

Which of the following is the contamination?

Salt water ▼

Which of the following is used as the primary treatment for this contamination?

Dilute system with water or swap to a salt saturated ▼

2 out of 2

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	14.0
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	12/21
API	6.0	11.1
Retort solids	17	17
Retort Oil/Water	0 / 83	0 / 83
MBT	2.5	2.5
pH	10.0	8.3
Pm	1.5	0.7
Pf/Mf	1.0/1.8	2.2/7.4
Chlorides	300	300
Hardness	100	0

Which of the following is the contamination?

Bicarbonates ▾

Which of the following is used as the primary treatment for this contamination?

Lime ▾

Which is the concentration of the above treatment?

1.5738 ▾

lb/bbl

3 out of 3

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	123
Weight	14.0	14.1
Vis	45	60
PV	30	45
YP	10	18
Gels	6/10	13/29
API	6.0	28.8
Retort solids	17	20
Retort Oil/Water	0 / 83	0 / 80
MBT	2.5	2.5
pH	10.0	12.1
Pm	1.5	6.7
Pf/Mf	1.0/1.8	4.4/4.5
Chlorides	300	300
Hardness	100	800

Which of the following is the contamination?

Cement ▾

Which of the following is used as the primary treatment for this contamination?

Sodium bicarbonate ▾

Which is the concentration of the above treatment?

0.5138 ▾

lb/bbl

3 out of 3

Question

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Given the following results:

	Day 1	Day 2
FL Temp	120	125
Weight	14.0	13.6
Funnel Vis	45	40
PV	30	24
YP	10	7
Gels	6/10	4/8
API	6.0	12.1
Retort solids	17	13
Retort Oil/Water	0 / 83	0 / 87
MBT	2.5	2.0
pH	10.0	9.3
Pm	1.5	1.0
Pf/Mf	1.0/1.1	0.8/0.9
Chlorides	1200	600
Hardness	100	80

Which of the following is the contamination?

Fresh water influx ▼

What is the first chemical that should be added to control this contamination?

Stop the influx weighting up ▼

2 out of 2

Question

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Calculate the concentration of alkalinity contributing ion(s) based on the given P_f and M_f results:

$P_f = 1.5$ ml $M_f = 3.7$ ml

OH, mg/L 0

CO₃, mg/L 1800

HCO₃, mg/L 854

3 out of 3

Question

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Calculate the concentration of alkalinity contributing ion(s) based on the given P_f and M_f results:

$P_f = 3.2$ ml $M_f = 3.2$ ml

OH, mg/L 1088

CO₃, mg/L 0

HCO₃, mg/L 0

3 out of 3
OH=1088
CO₃=0
HCO₃=0

Question

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Calculate the concentration of alkalinity contributing ion(s) based on the given P_f and M_f results:

P_f = 5.4 ml M_f = 7.1 ml

OH, mg/L 1258

CO₃, mg/L 2040

HCO₃, mg/L 0

3 out of 3

OH= 1258

CO₃= 2040

HCO₃= 0

Question

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Calculate the concentration of alkalinity contributing ion(s) based on the given P_f and M_f results:

P_f = 2.8 ml M_f = 5.6 ml

OH, mg/L 0

CO₃, mg/L 3360

HCO₃, mg/L 0

3 out of 3

OH=0

CO₃=3360

HCO₃= 0

Question

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Calculate the concentration of alkalinity contributing ion(s) based on the given P_f and M_f results:

P_f = 0 ml M_f = 2.5 ml

OH, mg/L 0

CO₃, mg/L 0

HCO₃, mg/L 3050

3 out of 3

OH=0
CO3=0
HCO3=3050